DÉMO 10 · CHANGEMENTS ENTRE LES ÉTAPES

Cycle de vie et optimisations

Chaque étape comparée à la précédente : ce qui est ajouté et ce qui est retiré.

`+ ligne ajoutée` `− ligne retirée` `@@ position dans le fichier`

1. **Étape 1 → 2** · etape-2-react-memo (3 fichiers)
2. **Étape 2 → 3** · etape-3-usecallback-usememo (3 fichiers)

Étape 1 → 2 etape-1-cycle-logs → etape-2-react-memo

React.memo et le state seeMore pour observer les renders inutiles

COMPARÉ **src/components/Cars.tsx → src/components/CarsMemo.tsx** +12 −2

```
@@ -1,3 +1,4 @@
+ import React from "react";
  import { useEffect, useRef, useState } from "react";

  function Car({ make, color }) {
@@ -24,6 +25,8 @@
    );
  }

+ const MemoizedCar = React.memo(Car);
+
  const allCars = [
    { make: "BMW", color: "Blue" },
    { make: "Audi", color: "White" },
@@ -36,6 +39,7 @@

    const [cars, setCars] = useState<any>([]);
    const [loading, setLoading] = useState<boolean>(true);
+   const [seeMore, setSeeMore] = useState<boolean>(false);

    console.log(`Home Rendered ${renderCount.current} times`, {
      loading,
@@ -71,7 +75,7 @@

    return (
      <div>
-       <h1>Welcome to the Home Page</h1>
+       <h1>{"Welcome to the Home Page"}</h1>
        <h2>Car List</h2>

        {loading ? (
@@ -80,10 +84,16 @@
          <>
            <div className="container">
              {cars.map((car: any, index: number) => (
-               <Car key={index} make={car.make} color={car.color} />
+               <MemoizedCar key={index} make={car.make} color={car.color} />
              ))}
            </div>
            <button onClick={() => handleFilter("BMW")}>Set Make to BMW</button>
+           <button
+             style={{ margin: "10px" }}
```

```
+              onClick={() => setSeeMore(!seeMore)}
+            >
+              See More
+            </button>
          </>
        )}
      </div>
```

MODIFIÉ **src/App.jsx** +1 −1

```
@@ -13,7 +13,7 @@
        <button onClick={() => setShowHome((v) => !v)}>
          {showHome ? "Unmount Home" : "Mount Home"}
        </button>
-       {showHome ? <Home /> : <div>Home is hidden</div>}
+       {showHome ? <CarsMemo /> : <div>Home is hidden</div>}
      </div>
    );
  }
```

UTILISÉ **src/components/CarsMemo.tsx** +103 −0

```
@@ -1,0 +1,103 @@
+ import React from "react";
+ import { useEffect, useRef, useState } from "react";
+
+ function Car({ make, color }) {
+   const renderCount = useRef(0);
+   renderCount.current += 1;
+   let colorInfo = color ? color : "Unknown";
+
+   console.log(`Car ${make} Rendered ${renderCount.current} times`, {
+     make,
+   });
+
+   useEffect(() => {
+     console.log(`Car ${make} Mounted`);
+     return () => {
+       console.log(`Car ${make} UnMounted`);
+     };
+   }, [make]);
+
+   return (
+     <div className="car-item">
+       <div>{make}</div>
+       <div>{colorInfo}</div>
+     </div>
+   );
+ }
+
+ const MemoizedCar = React.memo(Car);
+
+ const allCars = [
+   { make: "BMW", color: "Blue" },
+   { make: "Audi", color: "White" },
+   { make: "Mercedes", color: "Black" },
+ ];
+
+ function Cars() {
+   const renderCount = useRef(0);
+   renderCount.current += 1;
+
+   const [cars, setCars] = useState<any>([]);
+   const [loading, setLoading] = useState<boolean>(true);
+   const [seeMore, setSeeMore] = useState<boolean>(false);
```

```
+
+   console.log(`Home Rendered ${renderCount.current} times`, {
+     loading,
+     cars: cars.length,
+   });
+
+   useEffect(() => {
+     console.log("Home Mounted");
+     console.log("Fetching cars...");
+     setLoading(true);
+
+     // const timerId =
+     setTimeout(() => {
+       console.log("Cars fetched");
+       setCars(allCars);
+       setLoading(false);
+     }, 1000);
+     return () => {
+       console.log("Home UnMounted");
+     };
+   }, []);
+
+   const handleFilter = (selectedMake: string) => {
+     console.log(`Filtering cars by make: ${selectedMake}`);
+     if (selectedMake === "ALL") {
+       setCars(allCars);
+     } else {
+       const filteredCars = allCars.filter((car) => car.make === selectedMake);
+       setCars(filteredCars);
+       console.log(`Home filtered by make ${selectedMake}`);
+     }
+   };
+
+   return (
+     <div>
+       <h1>{"Welcome to the Home Page"}</h1>
+       <h2>Car List</h2>
+
+       {loading ? (
+         <div className="loading">Loading cars...</div>
+       ) : (
+         <>
+           <div className="container">
+             {cars.map((car: any, index: number) => (
+               <MemoizedCar key={index} make={car.make} color={car.color} />
+             ))}
+           </div>
+           <button onClick={() => handleFilter("BMW")}>Set Make to BMW</button>
+           <button
+             style={{ margin: "10px" }}
+             onClick={() => setSeeMore(!seeMore)}
+           >
+             See More
+           </button>
+         </>
+       )}
+     </div>
+   );
+ }
+
+ export default Cars;
```

Étape 2 → 3 etape-2-react-memo → etape-3-usecallback-usememo

useCallback (onSelect) et useMemo (carStyle) pour stabiliser les props

COMPARÉ **src/components/CarsMemo.tsx → src/components/CarsUseMemo.tsx** +38 −12

```
@@ -1,31 +1,35 @@
- import React from "react";
+ import React, { useCallback, useMemo } from "react";
  import { useEffect, useRef, useState } from "react";

- function Car({ make, color }) {
+ function Car({ make, color, onSelect, style }: any) {
    const renderCount = useRef(0);
    renderCount.current += 1;
-   let colorInfo = color ? color : "Unknown";

-   console.log(`Car ${make} Rendered ${renderCount.current} times`, {
-     make,
-   });
+   console.log(`Car ${make} Rendered ${renderCount.current} times`);

    useEffect(() => {
      console.log(`Car ${make} Mounted`);
-     return () => {
-       console.log(`Car ${make} UnMounted`);
-     };
+     return () => console.log(`Car ${make} UnMounted`);
    }, [make]);

    return (
-     <div className="car-item">
+     <div className="car-item" style={style} onClick={() => onSelect(make)}>
        <div>{make}</div>
-       <div>{colorInfo}</div>
+       <div>{color ?? "Unknown"}</div>
      </div>
    );
  }

  const MemoizedCar = React.memo(Car);
+
+ /*
+ const MemoizedCar = React.memo(Car, (prevProps, nextProps) => {
+   return (
+     prevProps.brand === nextProps.brand &&
+     prevProps.color === nextProps.color
+   );
+ });
+ */

  const allCars = [
    { make: "BMW", color: "Blue" },
@@ -45,6 +49,22 @@
      loading,
      cars: cars.length,
    });
+
+   // useCallback: même référence entre les renders
+   const onSelect = useCallback((make: string) => {
+     console.log("Selected:", make);
+   }, []);
+
```

```
        58  +   // useMemo: même objet entre les renders
        59  +   const carStyle = useMemo(
        60  +     () => ({
        61  +       padding: "10px",
        62  +       marginBottom: "10px",
        63  +       border: "1px solid #ccc",
        64  +       cursor: "pointer",
        65  +     }),
        66  +     []
        67  +   );
48      68
49      69        useEffect(() => {
50      70          console.log("Home Mounted");
                @@ -84,7 +104,13 @@
84      104             <>
85      105               <div className="container">
86      106                 {cars.map((car: any, index: number) => (
87          -               <MemoizedCar key={index} make={car.make} color={car.color} />
        107 +               <MemoizedCar
        108 +                 key={index}
        109 +                 make={car.make}
        110 +                 color={car.color}
        111 +                 onSelect={onSelect}
        112 +                 style={carStyle}
        113 +               />
88      114                 ))}
89      115               </div>
90      116               <button onClick={() => handleFilter("BMW")}>Set Make to BMW</button>
```

MODIFIÉ **src/App.jsx** +1 −1

```
                @@ -13,7 +13,7 @@
13      13          <button onClick={() => setShowHome((v) => !v)}>
14      14            {showHome ? "Unmount Home" : "Mount Home"}
15      15          </button>
16          -       {showHome ? <CarsMemo /> : <div>Home is hidden</div>}
        16  +       {showHome ? <CarsUseMemo /> : <div>Home is hidden</div>}
17      17        </div>
18      18      );
19      19    }
```

UTILISÉ **src/components/CarsUseMemo.tsx** +129 −0

```
                @@ -1,0 +1,129 @@
        1   + import React, { useCallback, useMemo } from "react";
        2   + import { useEffect, useRef, useState } from "react";
        3   +
        4   + function Car({ make, color, onSelect, style }: any) {
        5   +   const renderCount = useRef(0);
        6   +   renderCount.current += 1;
        7   +
        8   +   console.log(`Car ${make} Rendered ${renderCount.current} times`);
        9   +
        10  +   useEffect(() => {
        11  +     console.log(`Car ${make} Mounted`);
        12  +     return () => console.log(`Car ${make} UnMounted`);
        13  +   }, [make]);
        14  +
        15  +   return (
        16  +     <div className="car-item" style={style} onClick={() => onSelect(make)}>
        17  +       <div>{make}</div>
        18  +       <div>{color ?? "Unknown"}</div>
        19  +     </div>
        20  +   );
```

```
+ }
+
+ const MemoizedCar = React.memo(Car);
+
+ /*
+ const MemoizedCar = React.memo(Car, (prevProps, nextProps) => {
+   return (
+     prevProps.brand === nextProps.brand &&
+     prevProps.color === nextProps.color
+   );
+ });
+ */
+
+ const allCars = [
+   { make: "BMW", color: "Blue" },
+   { make: "Audi", color: "White" },
+   { make: "Mercedes", color: "Black" },
+ ];
+
+ function Cars() {
+   const renderCount = useRef(0);
+   renderCount.current += 1;
+
+   const [cars, setCars] = useState<any>([]);
+   const [loading, setLoading] = useState<boolean>(true);
+   const [seeMore, setSeeMore] = useState<boolean>(false);
+
+   console.log(`Home Rendered ${renderCount.current} times`, {
+     loading,
+     cars: cars.length,
+   });
+
+   // useCallback: même référence entre les renders
+   const onSelect = useCallback((make: string) => {
+     console.log("Selected:", make);
+   }, []);
+
+   // useMemo: même objet entre les renders
+   const carStyle = useMemo(
+     () => ({
+       padding: "10px",
+       marginBottom: "10px",
+       border: "1px solid #ccc",
+       cursor: "pointer",
+     }),
+     []
+   );
+
+   useEffect(() => {
+     console.log("Home Mounted");
+     console.log("Fetching cars...");
+     setLoading(true);
+
+     // const timerId =
+     setTimeout(() => {
+       console.log("Cars fetched");
+       setCars(allCars);
+       setLoading(false);
+     }, 1000);
+     return () => {
+       console.log("Home UnMounted");
+     };
+   }, []);
+
+   const handleFilter = (selectedMake: string) => {
```

```
+     console.log(`Filtering cars by make: ${selectedMake}`);
+     if (selectedMake === "ALL") {
+       setCars(allCars);
+     } else {
+       const filteredCars = allCars.filter((car) => car.make === selectedMake);
+       setCars(filteredCars);
+       console.log(`Home filtered by make ${selectedMake}`);
+     }
+   };
+
+   return (
+     <div>
+       <h1>{"Welcome to the Home Page"}</h1>
+       <h2>Car List</h2>
+
+       {loading ? (
+         <div className="loading">Loading cars...</div>
+       ) : (
+         <>
+           <div className="container">
+             {cars.map((car: any, index: number) => (
+               <MemoizedCar
+                 key={index}
+                 make={car.make}
+                 color={car.color}
+                 onSelect={onSelect}
+                 style={carStyle}
+               />
+             ))}
+           </div>
+           <button onClick={() => handleFilter("BMW")}>Set Make to BMW</button>
+           <button
+             style={{ margin: "10px" }}
+             onClick={() => setSeeMore(!seeMore)}
+           >
+             See More
+           </button>
+         </>
+       )}
+     </div>
+   );
+ }
+
+ export default Cars;
```